

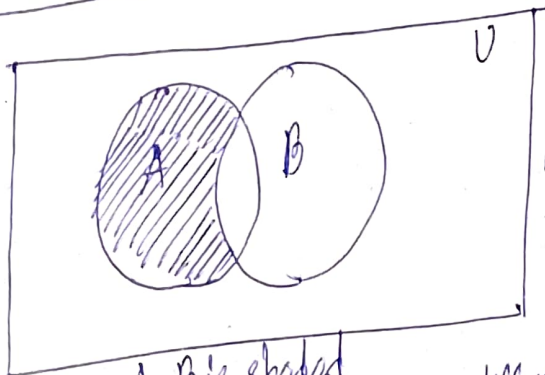
• We are often interested in finding the cardinality of a union of two finite sets A and B . Note that $|A| + |B|$ counts each element that is in A but not in B or in B but not in A exactly once, and each element that is in both A and B exactly twice.

• Thus, if the number of elements that are in both A and B is subtracted from $|A| + |B|$, elements in $A \cap B$ will be counted only once.

$$\therefore |A \cup B| = |A| + |B| - |A \cap B|$$

Definition 4: Let A and B be sets. The **difference** of A and B , denoted by $A - B$ or $A \setminus B$, is the set containing those elements that are in A but not in B . The difference of A and B is also called the **complement of B with respect to A** .

$$A - B = \{x \mid x \in A \wedge x \notin B\}$$



$A - B$ is shaded

Figure 3: Venn diagram for the difference of A and B

Example 6: $\{1, 3, 5\} - \{1, 2, 3\} = \{5\}$
 $\{1, 2, 3\} - \{1, 3, 5\} = \{2\}$ □

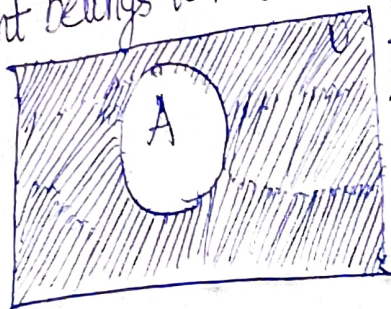
Example 7: The difference of the set of computer science majors at your school and the set of mathematics majors at your school is the set of all computer science majors at your school who are not also mathematics majors. □

mathematics majors. □

Definition 5: Let U be the universal set. The **complement** of the set A , denoted by $\bar{A}$, is the complement of A with respect to U . Therefore, the complement of the set A is $U - A$.

$$\bar{A} = \{x \in U \mid x \notin A\}$$

• An element belongs to $\bar{A}$ if and only if $x \notin A$.



$\bar{A}$ is shaded.

Figure 4: Venn diagram for the complement of the set A